

Problem Setup

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Problem Setup

Evaluate the indefinite integral

$$\int e^{-x^2} dx.$$

The integrand e^{-x^2} does not have an elementary antiderivative; we are asked to express the antiderivative in terms of a standard special function.

Strategy and Key Principles

The Gaussian integral

$$\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$$

leads to the definition of the error function

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Since the derivative of $\operatorname{erf}(x)$ reproduces the integrand e^{-x^2} up to a constant factor, we will rewrite the given integral in terms of $\operatorname{erf}(x)$.

Detailed Solution

First recall the definition of the error function:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Differentiate both sides with respect to x :

$$\frac{d}{dx} \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}.$$

Hence

$$e^{-x^2} = \frac{\sqrt{\pi}}{2} \frac{d}{dx} \operatorname{erf}(x).$$

Now integrate the original integrand:

$$\int e^{-x^2} dx = \int \frac{\sqrt{\pi}}{2} \frac{d}{dx} \operatorname{erf}(x) dx = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + C.$$

All steps are shown explicitly; the constant C denotes an arbitrary constant of integration.

Final Answer

$$\boxed{\int e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + C}$$

Key Insights

- The function e^{-x^2} has no elementary antiderivative; it is expressed via the error function $\operatorname{erf}(x)$.
- Differentiating the definition of $\operatorname{erf}(x)$ yields the exact factor needed to rewrite the integrand.
- A common mistake is to forget the factor $\frac{2}{\sqrt{\pi}}$ in the definition of $\operatorname{erf}(x)$, which leads to an incorrect constant in the final answer.